PocarLab Note
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Pump Speed Test — Report

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Version	Date	Description of changes	Editor
v0.1.0	2025-09-23	Initial pump speed test note	AA, GS
v0.2.0	2025-09-29	Corrected fits error bands	AA, GS

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This note documents a room-temperature pump characterization using water, intended to benchmark against vendor specifications and to inform HFE-based operation. Flows are reported in $\mathrm{L\,s^{-1}}$ throughout; $\mathrm{L\,min^{-1}}$ values are given parenthetically where helpful.

1 Vendor Specifications

- **For HFE/Novec 7000–7200** (viscosity 0.39 mPa s to 39.0 mPa s): capacity range 0.5 L min^{-1} to 4.0 L min^{-1}; motor speed range 200 rpm to 2150 rpm. The upper limit is 4.0 L min^{-1} at ∼2150 rpm (∼71.7 Hz).
- **Conversion between frequency and rpm:** 4-pole induction motor ⇒ $30n$ [rpm] ≈ f [Hz].

Digitized vendor water points at $\Delta p \approx 0$ bar are used for fits; performance curve is shown in Fig. 1.

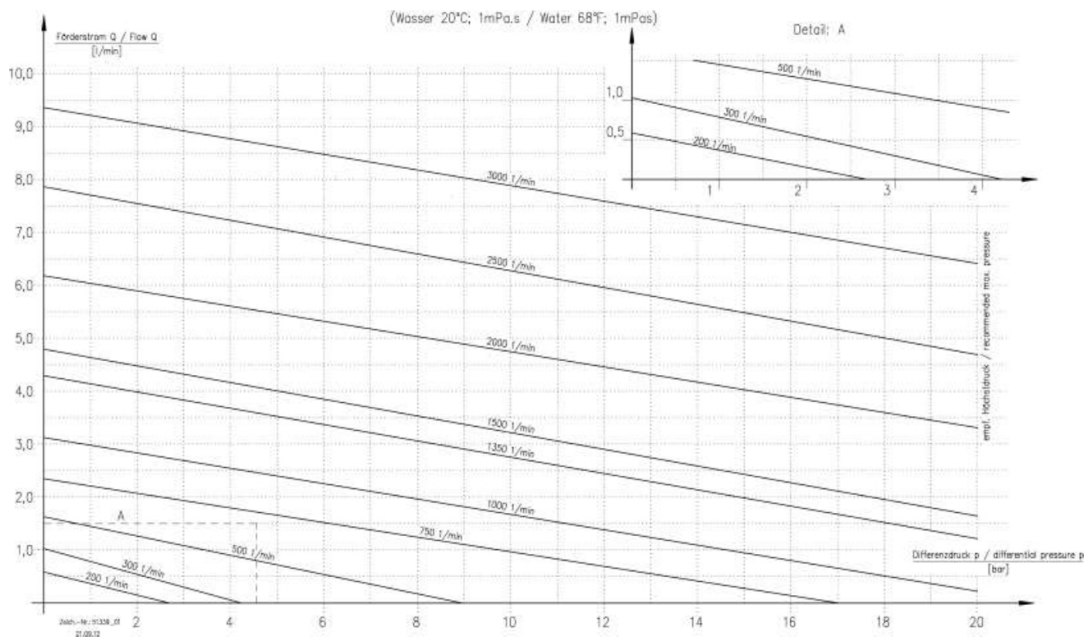


Figure 1: Vendor water performance curve.

Vendor water fit. A weighted linear fit forced through the origin yields

$$Q_{\text{vendor, water}}(f) \;=\; \big(0.00158 \pm 0.00008\big)\, f \quad [\mathrm{L\,s^{-1}}], \tag{1}$$

with f in Hz. The shaded band in plot 3 is an envelope obtained by refitting the same points as $y \pm \sigma$, where we assign $\sigma = 5\,\%$ with a 0.05 L min^{-1} floor. As a diagnostic, a free weighted fit $Q = af + b$ to the digitized points gives $b = -0.00026 \pm 0.00073$ L s^{-1} (0.36σ), i.e. statistically compatible with $b = 0$, so we report the physics-motivated $Q = mf$ form.

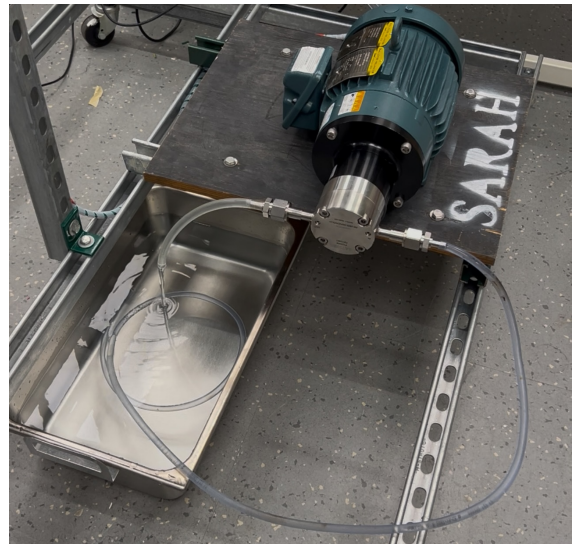
2 Measured Results (water, $\Delta p \approx 0$)

The pump was run at $f = 10$, 20, 30 and 40 Hz. Flow was measured volumetrically by timing the fill of a calibrated 500 mL container (±5 %); times were read to ±0.5 s. The outlet was open to atmosphere.

We use

$$Q = \frac{V}{t}, \qquad \sigma_Q^2 = \left(\frac{\sigma_V}{t}\right)^2 + \left(\frac{V}{t^2}\sigma_t\right)^2, \tag{2}$$

(a) Beaker setup.



(b) Pump setup.

Figure 2: Experimental test setup: (a) beaker and (b) pump.

with $V = 0.500 \pm 0.025$ L and $t \pm 0.5$ s.

Measurements are compiled in Table 1.

Frequency (Hz)	**Time to 0.5 L** (s)	**Flow** Q ($\mathrm{L\,s^{-1}}$)	$\pm 1\sigma$ ($\mathrm{L\,s^{-1}}$)	**Motor speed** (rpm)
10	28.82	0.0174	0.0010	∼300
20	14.52	0.0344	0.0022	∼600
30	9.77	0.0512	0.0037	∼900
40	7.37	0.0679	0.0056	∼1200

Table 1: Measured flows derived from a 0.5000 L volumetric fill. Uncertainties combine volume calibration and timing.

Measured water fit. A weighted linear fit forced through the origin yields

$$Q_{\text{measured}}(f) \;=\; \big(0.00172 \pm 0.00011\big)\, f \quad [\mathrm{L\,s^{-1}}], \tag{3}$$

with the band in plot 3 defined as the envelope from refitting the data as $y \pm \sigma$ using the propagated σ_Q. A free weighted fit $Q = af + b$ gives $b = 0.00044 \pm 0.00181$ $\mathrm{L\,s^{-1}}$ (0.24σ), so the measurement is statistically compatible with intersecting the origin; we therefore report the constrained $Q = mf$ result above.

3 HFE Specification

Two vendor points are available for the intended HFE working fluid: (200 rpm → 0.5 $\mathrm{L\,min^{-1}}$) and (2150 rpm → 4.0 $\mathrm{L\,min^{-1}}$).

HFE fit. A weighted linear fit through the origin yields

$$Q_{\text{HFE spec}}(f) \;=\; \big(0.00097 \pm 0.00006\big)\, f \quad [\mathrm{L\,s^{-1}}], \tag{4}$$

with the envelope band defined by refitting the two points as $y \pm \sigma$ using $\sigma = 5\,\%$ with a $0.05\,\mathrm{L\,min^{-1}}$ floor. As a diagnostic, a free fit $Q = af + b$ to these two points gives $b = 0.00235 \pm 0.00098\ \mathrm{L\,s^{-1}}$ (2.40σ), i.e. not statistically consistent with $b = 0$. Because the vendor does not specify behavior near $f = 0$ nor how the values for HFE were obtained, and on physical grounds, we report the origin-constrained result above.

4 Summary

All values and fits are shown in plot 3.

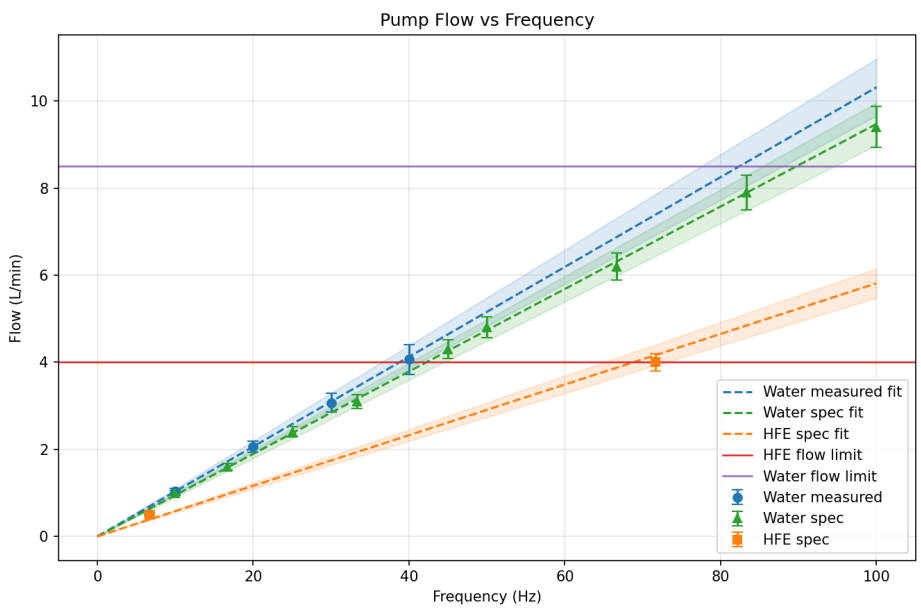


Figure 3: Measured water points with weighted fit and envelope band (blue), vendor water points with weighted fit and envelope band (green), and HFE spec points with weighted fit and envelope band (orange). Horizontal lines mark the HFE ($4\,\mathrm{L\,min^{-1}}$) and water ($\sim 8.5\,\mathrm{L\,min^{-1}}$) limits.

Our water calibration lies above the HFE/Novec specification, as expected for the lower viscosity of water, and scales consistently toward the motor's rated maximum.